

Datasheet Micro Servo TS90MD

Manufacturer: TiankongRC

Articel-Number: TS90MD – 270°

Link: [10PCS 9g Digital Micro Servo Full Metal Gear 90/180/270/360° PWM](#)



Figure 1: Micro Servo TS90MD

NOTE:

Please use a standard PWM control signal, the period is 20ms, the high-level time is 0.5-2.5ms. 270 degrees and 180 degrees can stop at any angle, and the angle can be adjusted. 360 degrees is a continuous rotation, and the angle and speed cannot be controlled.

180 degrees and 270 degrees can control the angle, PWM500-2500 remote control can only have half of the stroke.

(90 degree and 180 degree are the same servo)

SPECIFICATION:

Storage Temperature Range:	-30°C~80°C
Operating Temperature Range:	-25°C~70°C
Humidity Range:	65%±10%
Size:	22.7*12.2*26.8
Gear type:	5 Metal
Limit angle:	90/180/270/360°
Bearing:	No
Horn gear spline:	20T
Diameter:	4.8mm, Height:4.3mm
Horn type:	Plastic, POM
Case:	Engineering plastics(PBT)
Connector wire:	250mm±5mm
Motor:	Carbon brush motor
Splash water resistance:	No
Operat voltage:	4.8V - 6.0V
Idle current:	≤40mA
No load speed:	0.11sec/60° - 0.09sec/60°
Runnig current:	≤200mA - ≤300mA
Peak stall torque:	1.6kg.cm - 1.8kg.cm
Running torque:	1.2kg.cm/60° - 1.4kg.cm/60°
Stall current:	Max 580mA - Max800mA
Command signal:	Pulse width modification
Amplifier type:	Digital controller
Pulse width range:	500~2500usec
Neutral position:	1500usec
Running degree:	90/180/270°±3°(when 500~2500usec)
Running degree:	45/90/135°±3°(when 500~1500usec)
Running degree:	45/90/135°±3°(when 2500~1500usec)
Dead band width:	5 usec
Rotating direction:	Counterclockwise (when 1000~2000usec)

OUPTPUT PINOUT:

Red wire:	Positive +
Black wire:	Negative -
Brown wire:	Signal S